

Piracy Demand Measurement — System Comparison

Our pipeline (hash_trackerv2) vs the other team's DHT crawler · 2026-06-02 · INTERNAL

Verdict: Same core technique (prioritized iterative DHT crawl). **We lead on measurement completeness & product refinement; they lead on raw DHT scale & horizontal scalability.** Not strictly “better” — the two systems optimize different things.

Core method	DHT crawl + tracker-announce harvest (2 layers, unioned)	DHT crawl only	▲ US
DHT crawl scale	8 DHT workers, 1 box	60 processes, 2 machines (~750k resp/min)	▲ THEM
Measurement completeness	Union reaches full swarm magnitude (DHT alone undercounts 3–39×; harvester is the heavy lifter)	DHT-only = a sample ; structurally undercounts true size	▲ US
Trust / quality controls	txn validation · neighbor-spoof handling · node eviction · lock guards	txn validation · ID-shift detect · dup-list blacklist · bad-node timeout	= TIE
Infra / scalability	SQLite, single 4-core box (CPU-bound, SPOF)	Kafka + MySQL, 2 machines, horizontal scale-out	▲ THEM
Output / value	Per-country demand feed · Pantheon catalog ip_id · Movie/TV/Anime · velocity ranking · NBCU-validated · daily S3	Peers → MySQL · IP→country/ISP enrichment	▲ US
Resilience / ops	watchdog+SNS · auto WAL maintenance · md5-gated deploys (single box)	distributed buffer (Kafka) decouples crawl from storage	= TIE

■ **Numbers aren't directly comparable.** Their “~14M unique downloads/day” is a DHT-sample count; our feed (~6M distinct-IP rows/day) is **floored (≥10) and deduped per title×country×day**. Different metrics — no volume winner can be claimed from them. On raw DHT throughput, they are larger.

What would make us decisively better

- 1. Horizontal scale-out** — add DHT worker processes / a 2nd machine. Our DHT layer is the part that's smaller than theirs.
- 2. A buffer (Kafka-style) between crawl and storage** — decouples writers from the DB; would also eliminate the SQLite WAL contention we manage today.
- 3. Keep our two differentiators on top** — the tracker-harvest layer (true magnitude) and the demand-feed product (catalog-mapped, categorized, velocity, NBCU-validated).

Net: they have the better *scale architecture*; we have the better *method and product*. Our method + product on their scale-out beats either system alone.